

Writing Helper

Interruption-aware AI writing that learns a transparent style profile from stop points, repairs, and repeated revision choices.

CENTRAL CLAIM

Stop = Signal

A writer's interruption is not just a cancellation; it is evidence about voice, specificity, structure, and argument.

Introduction

Fluent AI prose can still feel wrong.

Writing Helper is a local browser prototype for interruption-aware multi-agent writing. When the user stops a generated sentence and chooses a better repair, the system treats that behavior as preference evidence.

Research question: Can interruption behavior recover latent preferences about wording, tone, structure, specificity, and argumentative style?

Motivation

Prompt-based personalization asks users to articulate their style in advance. This creates extra labor and favors strong prompt engineers.

This prototype learns from ordinary revision behavior: stop when a sentence feels wrong, choose or write a better option, and let repeated choices become inspectable memory.

Design goal: turn dissatisfaction into a durable, editable writing profile.

What It Contributes

- Behavior-first personalization from interruption and repair.
- Inspectable profile memory rather than hidden adaptation.
- Local profile storage, revision log, and audit export.
- Promotion rule: repeated evidence before durable memory.

Simulation Design

Each fake user receives a hidden profile drawn from wording, voice, style, and structure preferences. The assistant only observes interruptions and repairs, not the hidden profile itself.

A profile item is promoted only after three similar observations, which intentionally delays recall but reduces one-click overfitting.

Process Pipeline

Input Draft Stop Diagnose Options Choose Infer Memory Resume



user name + writing task Streaming Writer Agent user interrupts bad sentence Interpreter Agent Replacement Agent user selects option Behavior Interpreter Agent Preference Memory Agent profile-aware next draft

Each interruption creates a local observation; repeated observations are promoted into the global writing profile.

The pipeline begins with explicit user input, then alternates between agent work and user feedback. Each repair produces an observation that can later become profile memory.

Agents Involved

Streaming Writer drafts and resumes generation.

Interruption Interpreter explains why the stopped sentence may feel wrong.

Replacement Agent proposes candidate rewrites.

Behavior Interpreter interprets custom user feedback.

Preference Memory records local hints and promotes repeated preferences.

Example Trace

Interrupted draft: "Bioethics is a broad field that deals with medicine, technology, public health, and social values."

Chosen repair reason: Match latent style: avoid generic academic filler.

Memory update: observation 1/3; kept local until the same preference appears repeatedly.

Later promotion: after a third similar choice, "avoid generic academic filler" enters the global profile.

Data

100

simulated writers

30

steps each

2486

interruptions

512

manual actions

10.47

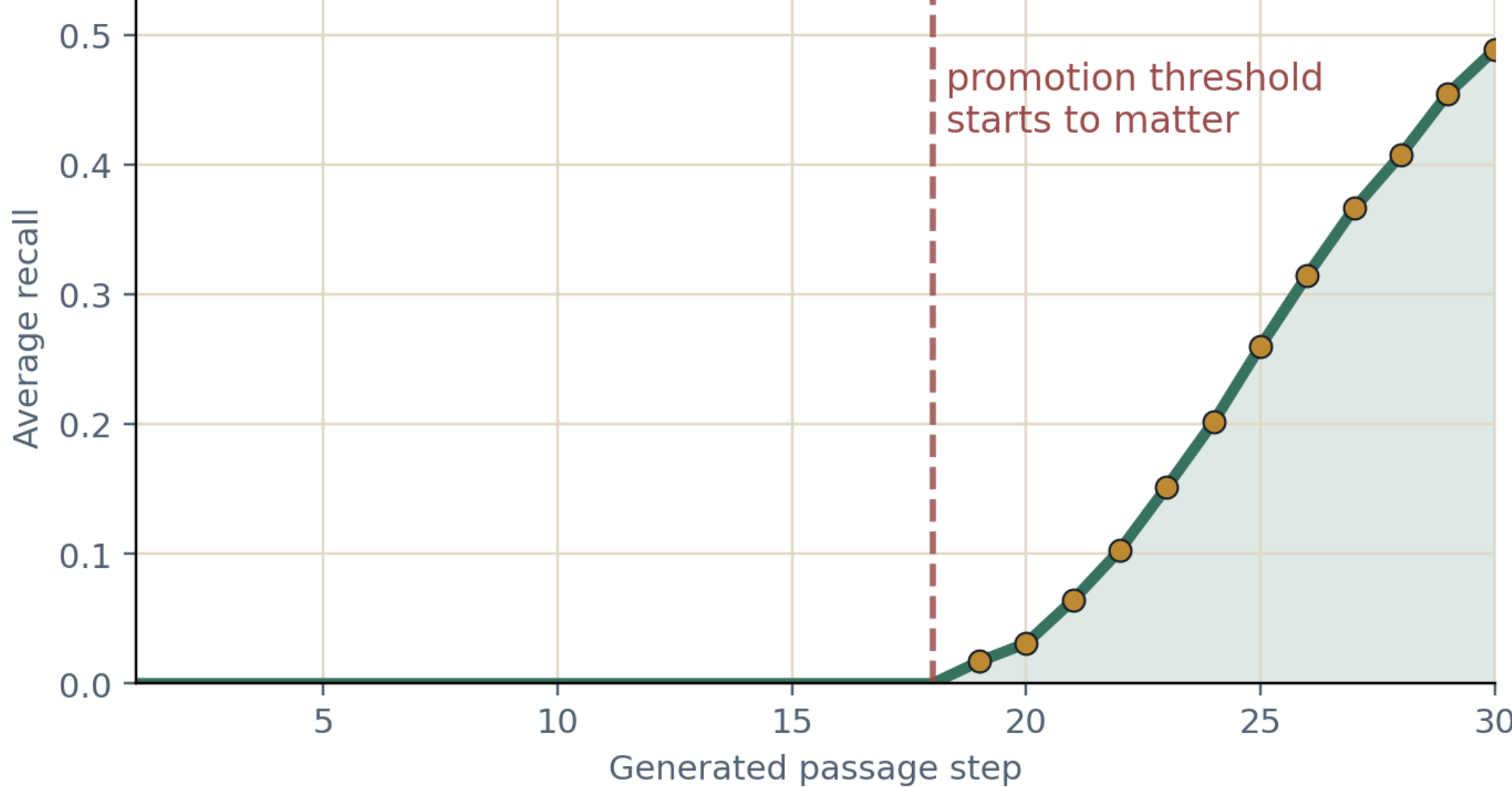
target items

5.00

recovered items

Results

Recovered Profile Recall Over 30 Writing Steps



0.489

mean final recall

0.495

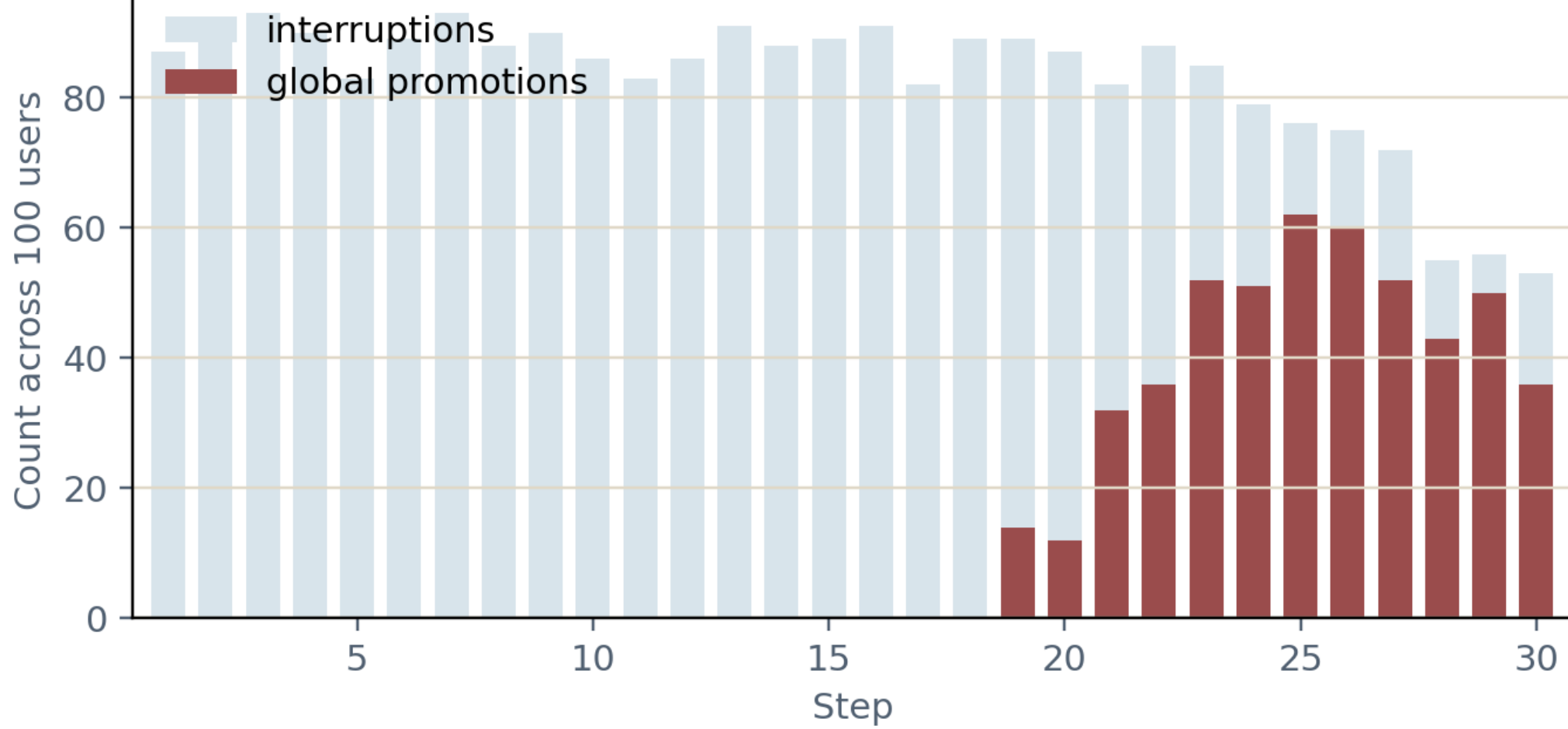
median final recall

0.923

max recall

Promotions

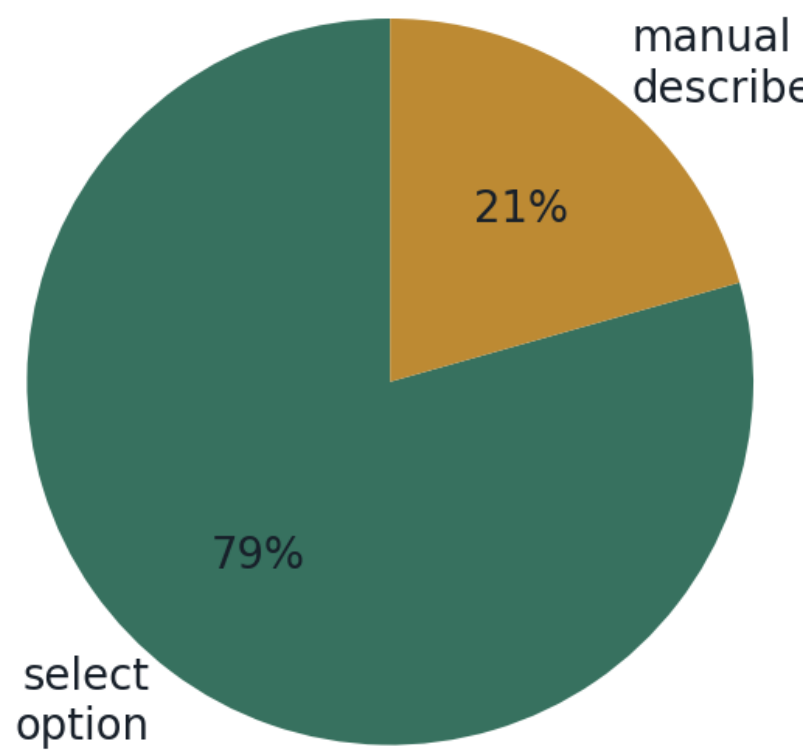
Interruptions vs. Durable Profile Promotions



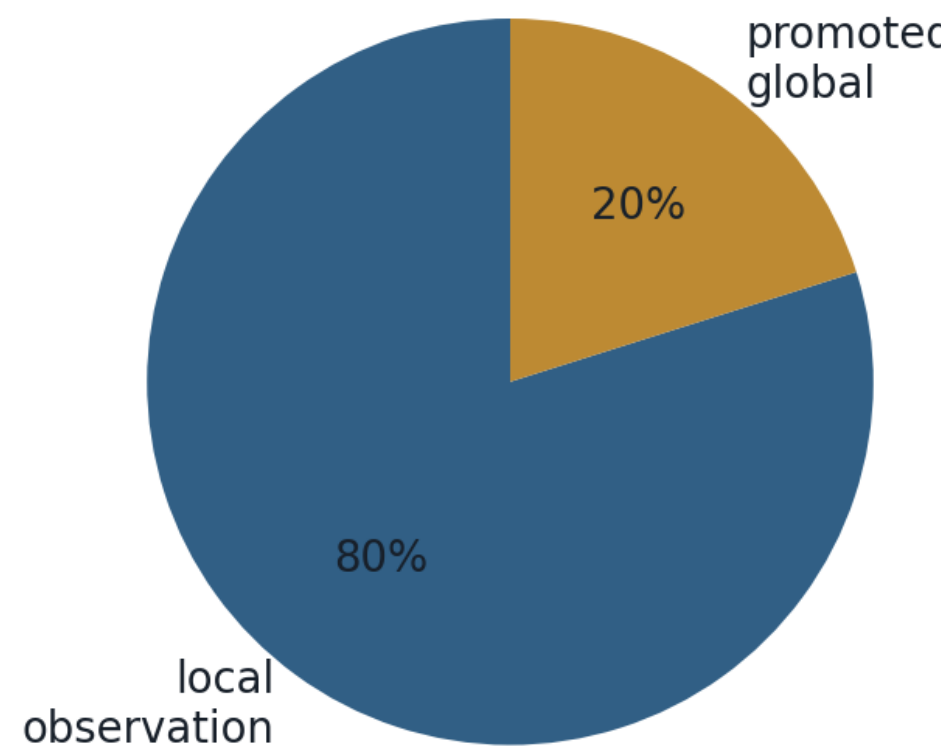
The system interrupts often, but durable profile updates appear later because a preference must recur before promotion.

Action Mix

Repair Action Mix



Memory Update Scope

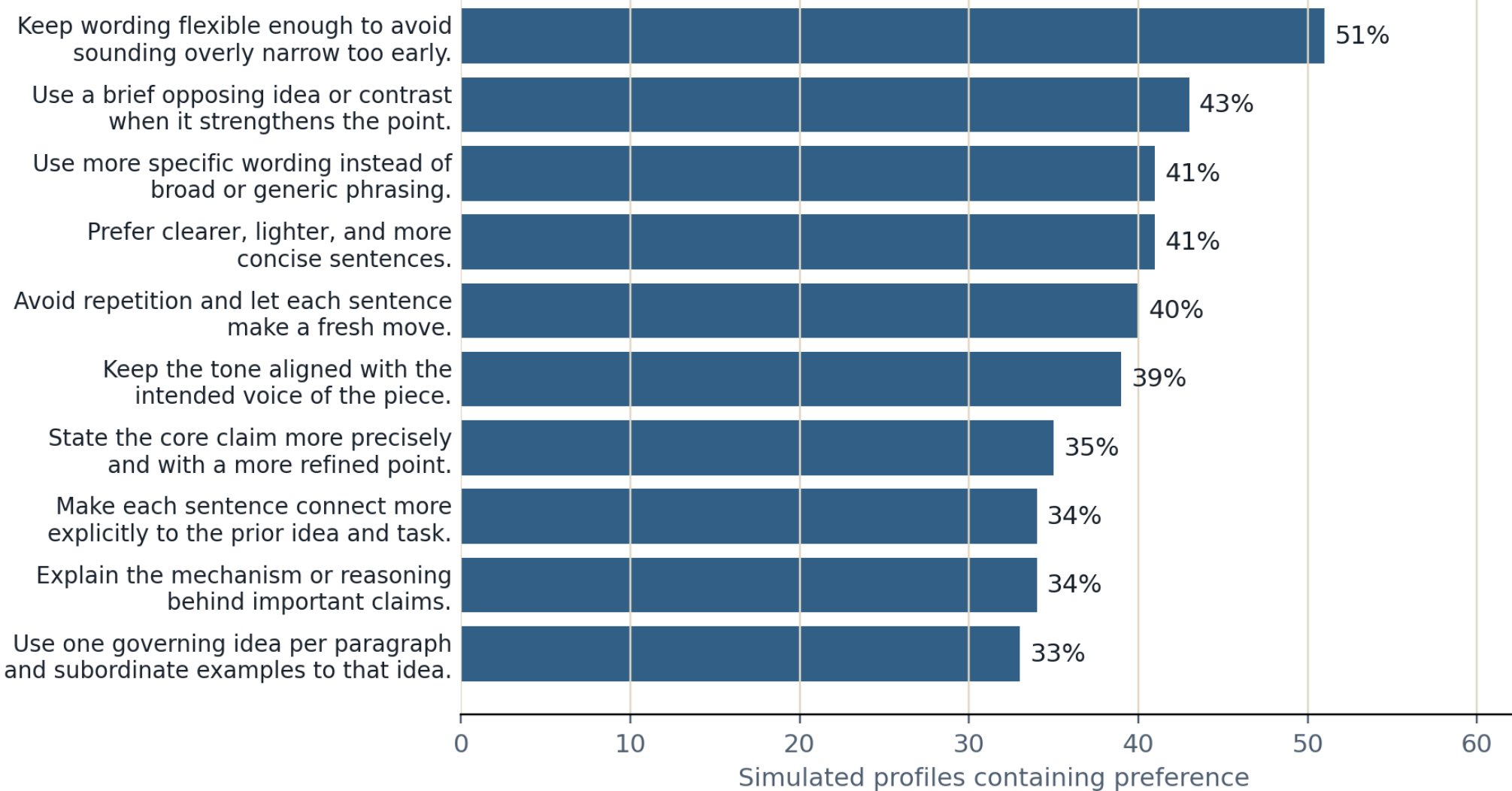


Result Reading

The curve is intentionally flat at first because the memory module requires repeated evidence. After step 18, repeated repairs begin converting local observations into global style memory.

Hidden Preferences

Most Frequent Hidden Writing Preferences



Interpretation

Low early recall is expected: one click is treated as weak local evidence.

The late rise shows that repeated stop-and-repair behavior can recover stable style preferences.

The run validates the logging and recovery pipeline, but it is still an offline simulation rather than a human study.

The next evaluation should compare recovered profiles against real writers' self-reported style preferences.

References Leading To This Work

CoAuthor. Lee, M., Liang, P., and Yang, Q. (2022). *CoAuthor: Designing a Human-AI Collaborative Writing Dataset for Exploring Language Model Capabilities*. The present work follows CoAuthor's insight that interaction traces, not only final text, reveal how people collaborate with language models. <https://arxiv.org/abs/2201.06796>

GhostWriter. Yeh, C., Ramos, G., Ng, R., Huntington, K., and Banks, R. (2024). *GhostWriter: Augmenting Collaborative Human-AI Writing Experiences Through Personalization and Agency*. Writing Helper narrows the personalization question to interruption points and selected repairs as style-learning evidence. <https://arxiv.org/abs/2402.08855>

Local prototype: streaming writer agent + interruption interpreter + replacement agent + transparent preference memory. Data source: deterministic offline recovery run in docs/poster_simulation.json.